

Foremost, Fastest, Shortest: Temporal Graph Realization under Various Path Metrics

Justine Cauvi, Nils Morawietz, Laurent Viennot

École Normale Supérieure de Lyon, Lyon, France

LaBRI, Université de Bordeaux, France, Institute of Computer Science,

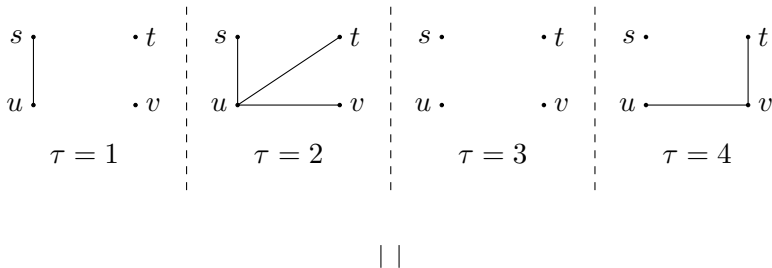
Friedrich Schiller University Jena, Germany

Inria, DI ENS, Paris, France

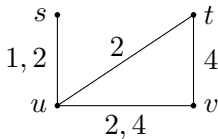
Meeting ANR Tempogral 2026 - Chamarande, France

Temporal graphs: models

Sequence of snapshots:

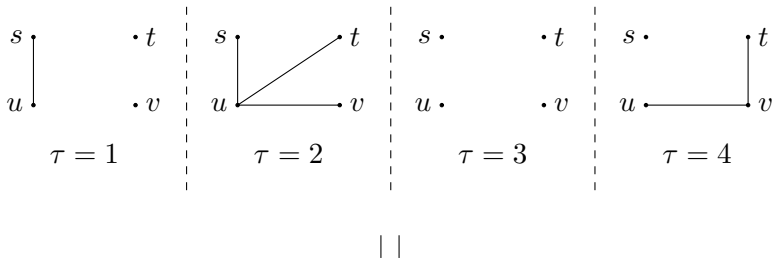


Labeled graph:

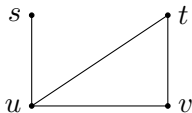


Temporal graphs: models

Sequence of snapshots:



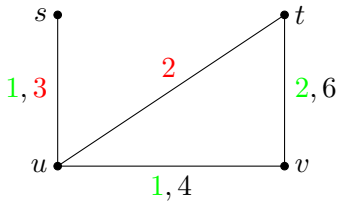
Labeled graph:



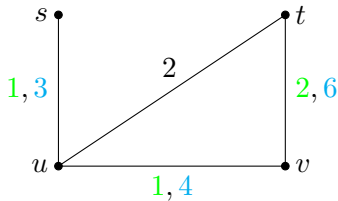
Underlying graph

Temporal paths: non strict and strict settings

Non-strict/Strict temporal path: A path and an **non-decreasing/increasing** sequence of times at which edges can be traversed one after the other



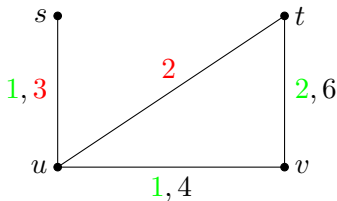
Non-strict setting



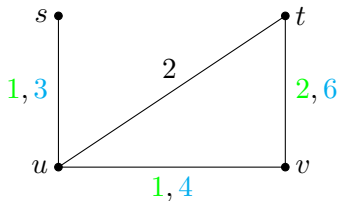
Strict setting

Temporal paths: non strict and strict settings

Non-strict/Strict temporal path: A path and an **non-decreasing/increasing** sequence of times at which edges can be traversed one after the other



Non-strict setting

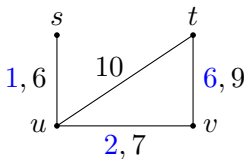


Strict setting

All the following examples are in the **strict setting**

Temporal paths: foremost, fastest, shortest

Temporal path: A path and an increasing sequence of times at which edges **can be traversed one after the other**

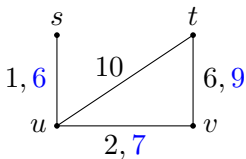


Several notions of distance:

- **Foremost** temporal path: earliest arrival time
- Fastest temporal path: minimum duration
- Shortest temporal path: minimum number of edges

Temporal paths: foremost, fastest, shortest

Temporal path: A path and an increasing sequence of times at which edges **can be traversed one after the other**

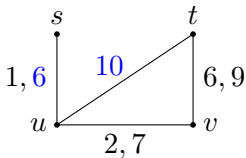


Several notions of distance:

- Foremost temporal path: earliest arrival time
- **Fastest** temporal path: minimum duration
- Shortest temporal path: minimum number of edges

Temporal paths: foremost, fastest, shortest

Temporal path: A path and an increasing sequence of times at which edges **can be traversed one after the other**



Several notions of distance:

- Foremost temporal path: earliest arrival time
- Fastest temporal path: minimum duration
- **Shortest** temporal path: minimum number of edges

Motivations and state of the art

Realization problems

Given a property P , is there a graph that satisfies the property?

Motivations and state of the art

Realization problems

Given a property P , is there a graph that satisfies the property?

State of the art:

PERIODIC FASTEST TEMPORAL GRAPH REALIZATION

Input: An $n \times n$ matrix D and a period Δ

Question: Is there a temporal graph with period Δ such that the minimum durations match the matrix?

Motivations and state of the art

Realization problems

Given a property P , is there a graph that satisfies the property?

State of the art:

PERIODIC FASTEST TEMPORAL GRAPH REALIZATION

Input: An $n \times n$ matrix D and a period Δ

Question: Is there a temporal graph with period Δ such that the minimum durations match the matrix?

- Railway networks, satellite networks
- **NP-hard** [Klobas et al. 2024]
- FPT by the feedback edge number of the underlying graph, $W[1]$ -hard by the feedback vertex number [Klobas et al. 2024]
- Generalized to up to ℓ labels per period [Erlebach et al. 2024]

Questions

- Does the hardness remains if we lift the periodic constraint and allow an unlimited number of labels?
- What if the matrix D gives the earliest arrival times? the minimum number of edges of a temporal path?

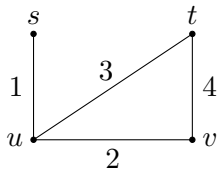
FOREMOST TGR

FOREMOST TGR

Input: An $n \times n$ matrix D with values in $\mathbb{N} \cup \{\infty\}$

Question: Is there a temporal graph G such that the earliest arrival times match the matrix?

$$\begin{array}{c} s \quad t \quad u \quad v \\ s \begin{pmatrix} 0 & 3 & 1 & 2 \end{pmatrix} \\ t \begin{pmatrix} \infty & 0 & 3 & 4 \end{pmatrix} \\ u \begin{pmatrix} 1 & 3 & 0 & 2 \end{pmatrix} \\ v \begin{pmatrix} \infty & 3 & 2 & 0 \end{pmatrix} \end{array}$$



$$\begin{array}{c} s \quad t \quad u \quad v \\ s \begin{pmatrix} 0 & \mathbf{3} & 1 & 2 \end{pmatrix} \\ t \begin{pmatrix} \mathbf{1} & 0 & 3 & 4 \end{pmatrix} \\ u \begin{pmatrix} 1 & 3 & 0 & 2 \end{pmatrix} \\ v \begin{pmatrix} \infty & 3 & 2 & 0 \end{pmatrix} \end{array}$$



NO

Our results

	Periodic	Non periodic
FOREMOST TGR	Polytime	Polytime
SHORTEST TGR	Polytime	NP-hard
FASTEST TGR	NP-hard	NP-hard

FOREMOST TGR

Theorem

FOREMOST TGR can be solved in $\mathcal{O}(n^3 \log n)$ time and $\mathcal{O}(n^2)$ space.

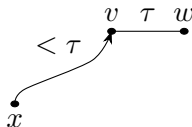
FOREMOST TGR

Theorem

FOREMOST TGR can be solved in $\mathcal{O}(n^3 \log n)$ time and $\mathcal{O}(n^2)$ space.

Edge compatibility

A temporal edge $(\{v, w\}, \tau)$ is **edge-compatible** with the matrix D if it satisfies: $EdgeCompat(D, \{v, w\}, \tau) := \forall x \in [n], D_{xv} < \tau \Rightarrow D_{xw} \leq \tau$ and $D_{xw} < \tau \Rightarrow D_{xv} \leq \tau$.



FOREMOST TGR

Theorem

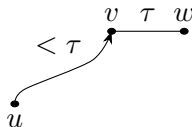
FOREMOST TGR can be solved in $\mathcal{O}(n^3 \log n)$ time and $\mathcal{O}(n^2)$ space.

Edge compatibility

A temporal edge $(\{v, w\}, \tau)$ is **edge-compatible** with the matrix D if it satisfies: $EdgeCompat(D, \{v, w\}, \tau) := \forall x \in [n], D_{xv} < \tau \Rightarrow D_{xw} \leq \tau$ and $D_{xw} < \tau \Rightarrow D_{xv} \leq \tau$.

Algorithm: For each entry $\tau = D_{uw}$, check that there exists v such that:

- $D_{uv} < \tau$,
- $(\{v, w\}, \tau)$ is edge-compatible.



Theorem

FOREMOST TGR can be solved in $\mathcal{O}(n^3 \log n)$ time and $\mathcal{O}(n^2)$ space.

Remarks:

- Produces a labeling with at most n labels per edge and at most n^2 labels in total: this is somehow tight

Theorem

FOREMOST TGR can be solved in $\mathcal{O}(n^3 \log n)$ time and $\mathcal{O}(n^2)$ space.

Remarks:

- Produces a labeling with at most n labels per edge and at most n^2 labels in total: this is somehow tight
- Valid for strict and non-strict settings, periodic graphs (slight change to the definition of edge-compatibility)

Theorem

FOREMOST TGR can be solved in $\mathcal{O}(n^3 \log n)$ time and $\mathcal{O}(n^2)$ space.

Remarks:

- Produces a labeling with at most n labels per edge and **at most n^2 labels** in total: this is somehow **tight**
- Valid for strict and non-strict settings, periodic graphs (slight change to the definition of edge-compatibility)
- Valid with a **prescribed graph** (if non-strict setting $\Rightarrow \mathcal{O}(n^2 m)$ with m the number of edges of the prescribed graph)

FOREMOST TGR: limitations of the tractability

Theorem

FOREMOST TGR is NP-hard when allowing at most one label per edge.

FOREMOST TGR: limitations of the tractability

Theorem

FOREMOST TGR is **NP-hard** when allowing **at most one label per edge**.

RANGED-FOREMOST TGR

Input: An $n \times n$ matrix D where each entry represents a range $[\ell, r]$

Question: Is there a temporal graph G such that the earliest arrival times match the matrix?

Theorem

- RANGED-FOREMOST TGR is **NP-hard**.

FOREMOST TGR: limitations of the tractability

Theorem

FOREMOST TGR is **NP-hard** when allowing **at most one label per edge**.

RANGED-FOREMOST TGR

Input: An $n \times n$ matrix D where each entry represents a range $[\ell, r]$

Question: Is there a temporal graph G such that the earliest arrival times match the matrix?

Theorem

- RANGED-FOREMOST TGR is **NP-hard**.
- RANGED-FOREMOST TGR can be solved in $\mathcal{O}(k^2 3^k n^4)$ time and $\mathcal{O}(2^k + n^2)$ space, where k is the **number of undetermined entries of D** .

General idea: Process by increasing time and guess which undetermined entries can be realized at the current time.

SHORTEST and FASTEST TGR

Theorem

SHORTEST TGR is NP-hard.

Remarks:

- Valid for strict and non-strict settings
- Polynomial-time solvable in periodic graphs (check if D is the distance matrix of the underlying graph)

SHORTEST and FASTEST TGR

Theorem

SHORTEST TGR is NP-hard.

Remarks:

- Valid for strict and non-strict settings
- Polynomial-time solvable in periodic graphs (check if D is the distance matrix of the underlying graph)

Theorem

FASTEST TGR is NP-hard and $W[1]$ -hard when parameterized by the vertex cover number of the underlying graph plus the largest entry of the matrix.

Remark: Valid for strict and non-strict settings, periodic graphs

Conclusion and future work

Summary:

	Periodic	Non periodic
FOREMOST TGR	Polytime	Polytime
SHORTEST TGR	Polytime	NP-hard
FASTEST TGR	NP-hard	NP-hard

Future work:

- Does SHORTEST TGR become polynomial-time solvable if we restrict the respective labeling?
- Are there structural parameters for which we can solve SHORTEST TGR in FPT-time? (e.g. treewidth)
- Are there constant factor approximations for FASTEST TGR and SHORTEST TGR? (Under the measurement of fulfilling as many entries as possible)

Conclusion and future work

Summary:

	Periodic	Non periodic
FOREMOST TGR	Polytime	Polytime
SHORTEST TGR	Polytime	NP-hard
FASTEST TGR	NP-hard	NP-hard

Future work:

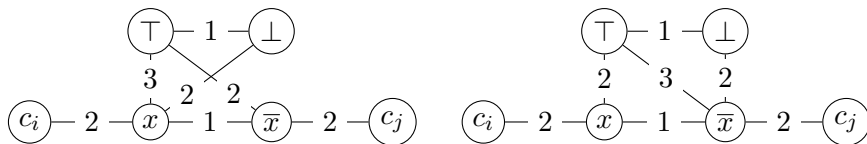
- Does SHORTEST TGR become polynomial-time solvable if we restrict the respective labeling?
- Are there structural parameters for which we can solve SHORTEST TGR in FPT-time? (e.g. treewidth)
- Are there constant factor approximations for FASTEST TGR and SHORTEST TGR? (Under the measurement of fulfilling as many entries as possible)

Thank you for listening!

References

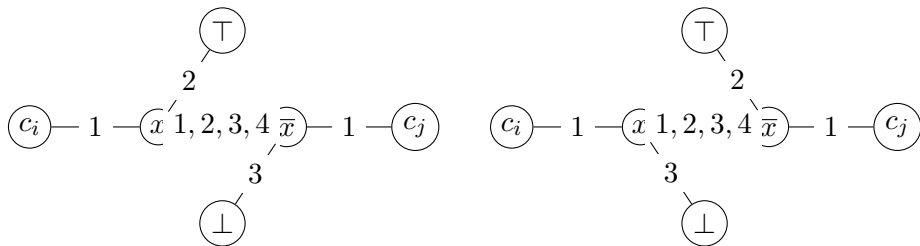
- [Klobas et al. 2024]: Nina Klobas, George B. Mertzios, Hendrik Molter, and Paul G. Spirakis. Temporal Graph Realization from Fastest Paths. In 3rd Symposium on Algorithmic Foundations of Dynamic Networks (SAND 2024).
- [Erlebach et al. 2024]: Thomas Erlebach, Nils Morawietz, and Petra Wolf. Parameterized Algorithms for Multi-Label Periodic Temporal Graph Realization. In 3rd Symposium on Algorithmic Foundations of Dynamic Networks (SAND 2024).

Foremost path with at most one label per edge



	c_i	c_j	x	$\bar{x}$	$\top$	$\perp$
c_i	0		2		3	
c_j		0		2	3	
x	2	2	0	1	2	2
$\bar{x}$	2	2	1	0	2	2
$\top$			2	2	0	1
$\perp$			2	2	1	0

NP-hardness with ranges



	c_i	c_j	x	$\bar{x}$	$\top$	$\perp$
c_i	0		1	2	2	3
c_j		0	2	1	2	3
x	1		0	1	2	3
$\bar{x}$		1	1	0	2	3
$\top$			[2,3]	[2,3]	0	
$\perp$			[3,4]	[3,4]		0

NP-hardness for shortest

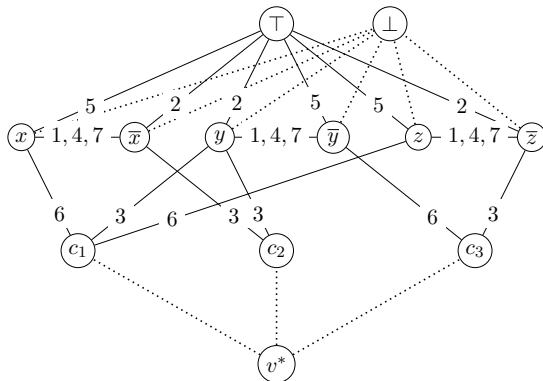


Figure: An example of the reduction for the formula $(x \vee y \vee z) \wedge (\bar{x} \vee y) \wedge (\bar{y} \vee \bar{z})$. A labeling that realizes the matrix D is depicted, where the dashed arcs receive the label set $\{1, 2, 7, 8\}$. Moreover this labeling corresponds to a satisfying truth assignment ($x = \text{False}$, $y = \text{True}$, $z = \text{False}$).

NP-hardness for fastest

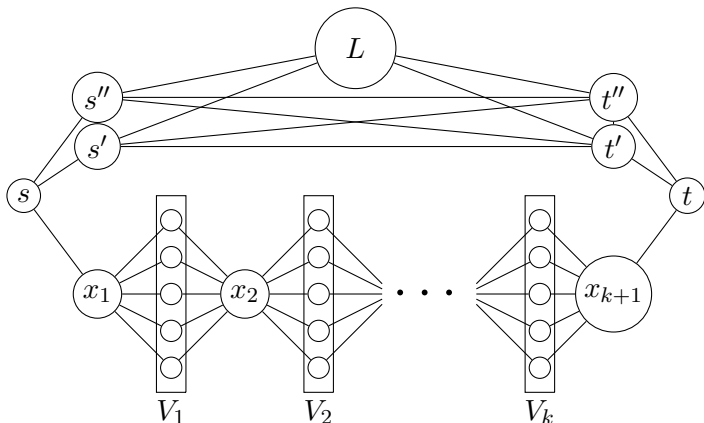


Figure: An illustration of the underlying graph from the reduction. The edges of the biclique $(L \cup \{s', s'', t', t''\}, V \cup X)$ are not depicted. Note that L is a clique of size $2k + 2$.